

Intake of Small-to-Medium-Chain Saturated Fatty Acids Is Associated with Peripheral Leukocyte Telomere Length in Postmenopausal Women

Dietary factors, including dietary fat, may affect the biological aging process, as reflected by the shortening of telomere length (TL), by affecting levels of oxidative stress and inflammatory responses. We examined the direct relations of total and types of dietary fats and fat-rich foods to peripheral leukocyte TL. In 4029 apparently healthy postmenopausal women who participated in the Women's Health Initiative, intakes of total fat, individual fatty acids, and fat-rich foods were assessed by a questionnaire. TL was measured by quantitative polymerase chain reaction. Intake of short-to-medium-chain saturated fatty acids (SMSFAs; aliphatic tails of ≤ 12 carbons) was inversely associated with TL. Compared with participants in other quartiles of SMSFA intake, women who were in the highest quartile (median: 1.29% of energy) had shorter TLs [mean: 4.00 kb (95% CI: 3.89, 4.11 kb)], whereas women in the lowest quartile of intake (median: 0.29% of energy) had longer TLs [mean: 4.13 kb (95% CI: 4.03, 4.24 kb); P-trend = 0.046]. Except for lauric acid, all other individual SMSFAs were inversely associated with TL ($P < 0.05$). In isoenergetic substitution models, the substitution of 1% of energy from SMSFAs with any other energy source was associated with 119 bp longer TLs (95% CI: 21, 216 bp). Intakes of nonskim milk, butter, and whole-milk cheese (major sources of SMSFAs) were all inversely associated with TL. No significant associations were found with long-chain saturated fatty acids, monounsaturated fatty acids, and polyunsaturated fatty acids. In conclusion, we found that higher intakes of SMSFAs and SMSFA-rich foods were associated with shorter peripheral leukocyte TL among postmenopausal women. These findings suggest the potential roles of SMSFAs in the rate of biological aging.